ADVANCE COLLEGE OF ENGINEERING AND MANAGEMENT

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LAB NO: 2
Subject: AI

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Submitted To:
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Title: Implementation of BackTracking Algorithm

Objectives :

- To understand the concept and working principle of the Backtracking algorithm
- To implement backtracking for solving constraint-based problems
- To study the application of backtracking in Constraint Satisfaction Problems (CSP)
- To analyze adversarial problema scenarios using backtracking techniques

Theory

1. Backtracking Algorithm

Backtracking is a recursive problem-solving technique that builds solutions incrementally and abandons a solution as soon as it determines that the solution cannot be completed successfully. It systematically explores all possible configurations using a depth-first search approach. If a partial solution violates constraints, the algorithm backtracks to try a different path. This reduces unnecessary computations compared to brute-force methods.

2. Constraint Satisfaction Problem (CSP)

A Constraint Satisfaction Problem (CSP) is a problem where a set of variables must be assigned values from given domains in such a way that all specified constraints are satisfied. Each variable has a domain of possible values, and constraints define the rules that restrict the combinations of values. Examples of CSP include map coloring, Sudoku, and N-Queens problems. The goal is to find a solution that satisfies all constraints.

3. Backtracking in Adversarial Problems

In adversarial problems, decisions are made in the presence of opposing conditions or competing choices. Backtracking helps explore all possible decision paths and outcomes to ensure that the best possible solution is found. It is useful in scenarios where each choice can affect future possibilities, and all alternatives must be evaluated carefully. This approach is commonly applied in game strategies and decision-making problems.

Overall, backtracking is a powerful technique for solving LSP and adversarial problems, though its efficiency depends on the size of the input and the effectiveness of pruning techniques.

Backtracking Source-code

```
 import time
import tkinter as tk
from tkinter import messagebox


class TreeNode:
```

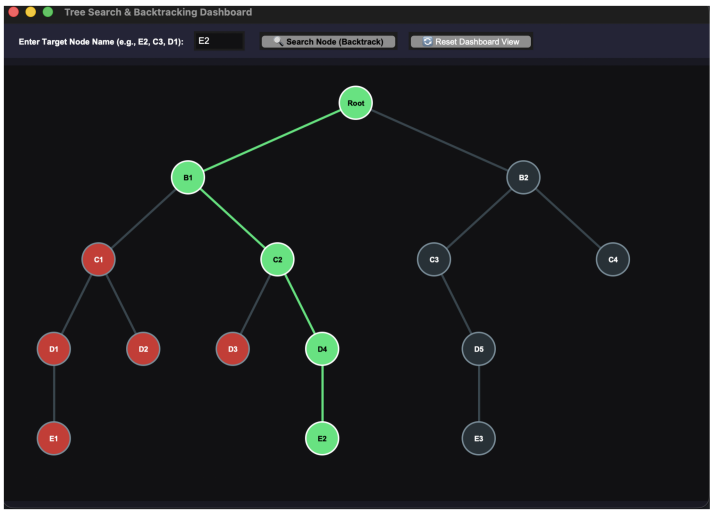
```
    def __init__(self, name, x, y):
        self.name = name          # Unique Identifier (e.g., 'A', 'B', 'C')
        self.x = x                # Canvas X coordinate
        self.y = y                # Canvas Y coordinate
        self.children = []        # Sub-nodes linked below


class TreeBacktrackDashboard:
    def __init__(self, root):
        self.root = root
        self.root.title("Tree Search & Backtracking Dashboard")
        self.root.geometry("950x650")
        self.root.configure(bg='#1A1A24')

        # Control States
        self.search_target = ""
        self.path_taken = []      # Active tracking stack framework
        self.visited_nodes = set()

        self.create_tree_structure()
        self.create_widgets()

    def create_tree_structure(self):
        """Constructs a deterministic tree with 15 nodes and a depth of 5."""
        # Layer 1: Root (Depth 1)
        self.root_node = TreeNode("Root", 475, 50)

        # Layer 2: (Depth 2)
        b1 = TreeNode("B1", 250, 150)
        b2 = TreeNode("B2", 700, 150)
        self.root_node.children = [b1, b2]

        # Layer 3: (Depth 3)
        c1 = TreeNode("C1", 130, 260)
        c2 = TreeNode("C2", 370, 260)
        c3 = TreeNode("C3", 580, 260)
        c4 = TreeNode("C4", 820, 260)
        b1.children = [c1, c2]
        b2.children = [c3, c4]

        # Layer 4: (Depth 4)
        d1 = TreeNode("D1", 70, 380)
        d2 = TreeNode("D2", 190, 380)
        d3 = TreeNode("D3", 310, 380)
        d4 = TreeNode("D4", 430, 380)
        d5 = TreeNode("D5", 640, 380)
        c1.children = [d1, d2]
        c2.children = [d3, d4]
        c3.children = [d5]  # Leaving asymmetry
```

```
        # Layer 5: (Depth 5)
        e1 = TreeNode("E1", 70, 500)
        e2 = TreeNode("E2", 430, 500)
        d1.children = [e1]
        d4.children = [e2]

        # Total Nodes count = 1(Root) + 2(B) + 4(C) + 5(D) + 2(E) = 14 nodes.
        # Let's add one more node to make it exactly 15 nodes total.
        e3 = TreeNode("E3", 640, 500)
        d5.children = [e3]

    def create_widgets(self):
        # --- Upper Control Bar ---
        control_frame = tk.Frame(self.root, bg='#252538', pady=10)
        control_frame.pack(side=tk.TOP, fill=tk.X)

        tk.Label(
            control_frame, text="Enter Target Node Name (e.g., E2, C3, D1):",
            font=('Arial', 11, 'bold'), bg='#252538', fg='white'
        ).pack(side=tk.LEFT, padx=(20, 5))

        self.entry_target = tk.Entry(control_frame, font=('Arial', 12), width=8,
bg='#121214', fg='white', insertbackground='white')
        self.entry_target.pack(side=tk.LEFT, padx=5)
        self.entry_target.insert(0, "E2") # Default search goal text

        search_btn = tk.Button(
            control_frame, text="�� Search Node (Backtrack)", font=('Arial', 11, 'bold'),
            bg='#00E676', fg='black', command=self.start_backtrack_search,
relief=tk.FLAT
        )
        search_btn.pack(side=tk.LEFT, padx=15)

        reset_btn = tk.Button(
            control_frame, text="�� Reset Dashboard View", font=('Arial', 11),
            bg='#37474F', fg='white', command=self.reset_display, relief=tk.FLAT
        )
        reset_btn.pack(side=tk.LEFT)

        # --- Tree Rendering Canvas ---
        self.canvas = tk.Canvas(self.root, bg='#121214', highlightthickness=0)
        self.canvas.pack(fill=tk.BOTH, expand=True, pady=10)

        self.draw_tree(self.root_node)

    def draw_tree(self, node):
        """Recursively renders connections and circle nodes to the graphics viewport."""
        for child in node.children:
            # Draw structural links
            color = '#37474F'
```

```
            # Highlight edge if both parent and child are on the active search path
            if node in self.path_taken and child in self.path_taken:
                color = '#00E676' # Neon green connection path
            elif node in self.visited_nodes and child in self.visited_nodes:
                color = '#D32F2F' # Dull crimson path for rejected configurations

            self.canvas.create_line(node.x, node.y, child.x, child.y, width=3, fill=color)
            self.draw_tree(child)

        # Render current node node body
        fill_color = '#263238'
        outline_color = '#78909C'

        if node in self.path_taken:
            fill_color = '#00E676'  # Green if actively on the solution path
            outline_color = '#FFFFFF'
        elif node.name in self.visited_nodes:
            fill_color = '#D32F2F'  # Red if it was a dead-end and we backtracked off it

        self.canvas.create_oval(node.x-22, node.y-22, node.x+22, node.y+22,
fill=fill_color, outline=outline_color, width=2)
        text_color = 'white' if fill_color != '#00E676' else 'black'
        self.canvas.create_text(node.x, node.y, text=node.name, font=('Arial', 10, 'bold'),
fill=text_color)

    # --- Core Backtracking Algorithm ---
    def backtrack_search(self, current_node):
        # 1. Update UI state to show node entry
        self.path_taken.append(current_node)
        self.visited_nodes.add(current_node.name)

        self.refresh_canvas_view(250) # short execution pause to animate search

        # Base Case / Goal Test Success Check
        if current_node.name.upper() == self.search_target:
            return True

        # 2. Iterate through choices (Child branches)
        for child in current_node.children:
            # Recurse down the branch
            if self.backtrack_search(child):
                return True # Success propagation upward

        # 3. BACKTRACK STEP: If children did not find the target, this path is a dead
end.
        # We pop the node off our active solution stack to step back up to the parent.
        self.path_taken.pop()
        self.refresh_canvas_view(250) # short execution pause to animate backtracking

        return False
```

```
    def start_backtrack_search(self):
        self.search_target = self.entry_target.get().strip().upper()
        self.path_taken = []
        self.visited_nodes.clear()

        if not self.search_target:
            messagebox.showwarning("Input Error", "Please provide a valid node key
target.")
            return

        found = self.backtrack_search(self.root_node)

        if found:
            path_str = " -> ".join([node.name for node in self.path_taken])
            messagebox.showinfo("Target Found!", f"Successfully located node
{self.search_target}!\n\nBacktracked Verification Path:\n{path_str}")
        else:
            messagebox.showerror("Not Found", f"Node '{self.search_target}' does not
exist inside this tree topology structure.")

    def refresh_canvas_view(self, ms_delay):
        self.canvas.delete("all")
        self.draw_tree(self.root_node)
        self.root.update()
        self.root.after(ms_delay)

    def reset_display(self):
        self.path_taken.clear()
        self.visited_nodes.clear()
        self.canvas.delete("all")
        self.draw_tree(self.root_node)


if __name__ == "__main__":
    main_window = tk.Tk()
    app = TreeBacktrackDashboard(main_window)
    main_window.mainloop()
```

Output:



CSP Source-code

```
import math
import random
import tkinter as tk
from tkinter import messagebox


class DynamicCSP10NodesGUI:
    def __init__(self, root):
        self.root = root
        self.root.title("Dynamic CSP Playground (10 Nodes Layout)")
        self.root.geometry("900x650")
        self.root.configure(bg='#1e1e24')

        # Colors available for selection (Palette)
        self.colors = ['#FF5252', '#4CAF50', '#2196F3', '#FFEB3B']  # Red, Green,
Blue, Yellow
        self.color_names = ['Red', 'Green', 'Blue', 'Yellow']
        self.selected_color = self.colors[0]

        # CSP Variables
        self.num_nodes = 10
        self.nodes = [f"N{i}" for i in range(1, self.num_nodes + 1)]
        self.coords = {}        # {'N1': (x, y)}
        self.topology = {}      # {'N1': ['N2', 'N3']}
        self.assignments = {}   # Active solutions: {'N1': '#FF5252'}

        self.create_widgets()
        self.generate_10_node_graph()

    def create_widgets(self):
        # --- Top Control Panel ---
        control_frame = tk.Frame(self.root, bg='#2a2a35', pady=10)
        control_frame.pack(side=tk.TOP, fill=tk.X)
```

```
        gen_btn = tk.Button(
            control_frame, text="🎲 Generate New 10-Node Map", font=('Arial', 11,
'bold'),
            bg='#9C27B0', fg='white', command=self.generate_10_node_graph,
relief=tk.FLAT
        )
        gen_btn.pack(side=tk.LEFT, padx=15)

        solve_btn = tk.Button(
            control_frame, text="🤖 AI Auto-Solve", font=('Arial', 11, 'bold'),
            bg='#4CAF50', fg='white', command=self.run_ai_solver, relief=tk.FLAT
        )
        solve_btn.pack(side=tk.LEFT, padx=5)

        clear_btn = tk.Button(
            control_frame, text="🧹 Clear Colors", font=('Arial', 11),
            bg='#555566', fg='white', command=self.clear_assignments, relief=tk.FLAT
        )
        clear_btn.pack(side=tk.LEFT, padx=5)

        # --- Side Color Selection Palette ---
        palette_frame = tk.Frame(self.root, bg='#2a2a35', width=150, padx=15,
pady=20)
        palette_frame.pack(side=tk.LEFT, fill=tk.Y)

        palette_label = tk.Label(
            palette_frame, text="SELECT COLOR", font=('Arial', 10, 'bold'),
            bg='#2a2a35', fg='#B0BEC5'
        )
        palette_label.pack(anchor=tk.W, pady=(0, 15))

        self.color_var = tk.StringVar(value=self.colors[0])
        for i, color in enumerate(self.colors):
            frame = tk.Frame(palette_frame, bg='#2a2a35', pady=6)
            frame.pack(fill=tk.X)

            rb = tk.Radiobutton(
                frame, text=self.color_names[i], variable=self.color_var,
                value=color, bg='#2a2a35', fg='white', selectcolor='#2a2a35',
                font=('Arial', 11), command=self.update_selected_color
            )
            rb.pack(side=tk.LEFT)

            canvas_indicator = tk.Canvas(frame, width=20, height=20, bg=color,
highlightthickness=0)
            canvas_indicator.pack(side=tk.RIGHT, padx=5)

        # --- Graphics View Canvas ---
        self.canvas = tk.Canvas(self.root, bg='#121214', highlightthickness=0)
```

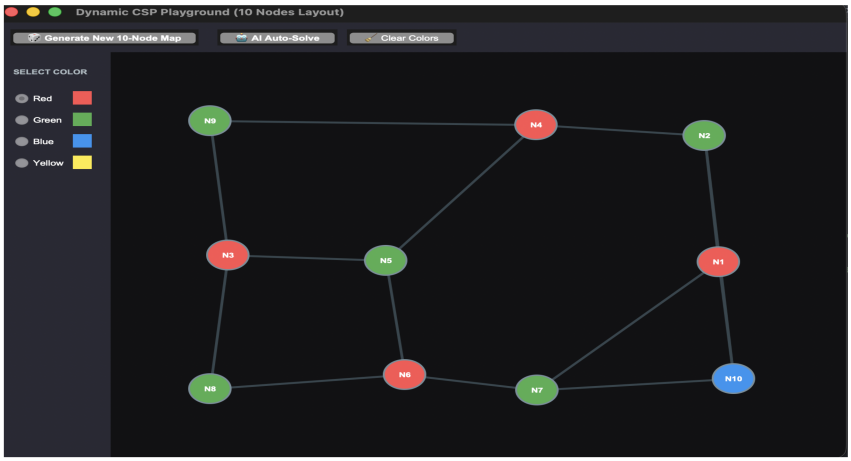
```
        self.canvas.pack(side=tk.RIGHT, fill=tk.BOTH, expand=True)
        self.canvas.bind("<Button-1>", self.handle_canvas_click)

    def generate_10_node_graph(self):
        """Generates a stable, non-overlapping 10-node layout using Grid Anchors."""
        self.assignments.clear()
        self.coords.clear()
        self.topology.clear()

        # Define 12 possible logical sectors on canvas (4 columns x 3 rows grid map)
        cols, rows = 4, 3
        cell_width = 700 // cols
        cell_height = 550 // rows

        sectors = [(c, r) for c in range(cols) for r in range(rows)]
        # Pick 10 random sectors out of the 12 to place our 10 nodes
        chosen_sectors = random.sample(sectors, self.num_nodes)

        # Generate coordinates with local jitter inside their sector bounds
        for idx, node in enumerate(self.nodes):
            sc_col, sc_row = chosen_sectors[idx]

            # Compute sector center coordinates
            base_x = int((sc_col + 0.5) * cell_width) + 30
            base_y = int((sc_row + 0.5) * cell_height) + 30

            # Inject a small random offset (jitter) to keep the layout feeling dynamic
            x = base_x + random.randint(-20, 20)
            y = base_y + random.randint(-20, 20)

            self.coords[node] = (x, y)
            self.topology[node] = []

        # Generate a realistic tree/mesh topology network
        for i, node in enumerate(self.nodes):
            # Sort all other nodes by physical distance from current node
            distances = [(target, math.hypot(self.coords[node][0] - self.coords[target][0],
                                             self.coords[node][1] - self.coords[target][1]))
                         for target in self.nodes if target != node]
            distances.sort(key=lambda item: item[1])

            # Force connect to closest neighbor to guarantee no node is isolated
            closest_neighbor = distances[0][0]
            if closest_neighbor not in self.topology[node]:
                self.topology[node].append(closest_neighbor)
                self.topology[closest_neighbor].append(node)

            # Randomly attach to secondary neighbors to build complex constraints
            num_extra_edges = random.randint(1, 2)
            for j in range(1, min(num_extra_edges + 1, len(distances))):
```

```
                candidate = distances[j][0]
                # Enforce max degree limits (max 3 edges per node) to maintain planar
colorability
                if len(self.topology[node]) < 3 and len(self.topology[candidate]) < 3:
                    if candidate not in self.topology[node]:
                        self.topology[node].append(candidate)
                        self.topology[candidate].append(node)

        self.draw_graph()

    def draw_graph(self):
        self.canvas.delete("all")

        # Draw constraints edges
        drawn_edges = set()
        for node, neighbors in self.topology.items():
            for neighbor in neighbors:
                edge = tuple(sorted((node, neighbor)))
                if edge not in drawn_edges:
                    x1, y1 = self.coords[node]
                    x2, y2 = self.coords[neighbor]
                    self.canvas.create_line(x1, y1, x2, y2, width=3, fill='#37474F')
                    drawn_edges.add(edge)

        # Draw structural node bodies
        for node, (x, y) in self.coords.items():
            color = self.assignments.get(node, '#3A3A4A')  # Dark gray if unassigned

            # Nodes rendering radius set to 22 pixels
            self.canvas.create_oval(x-22, y-22, x+22, y+22, fill=color,
outline='#78909C', width=2)
            self.canvas.create_text(x, y, text=node, font=('Arial', 10, 'bold'), fill='white')

    def update_selected_color(self):
        self.selected_color = self.color_var.get()

    def handle_canvas_click(self, event):
        """Processes human node interaction and triggers adjacency logic
confirmation checks."""
        clicked_node = None
        for node, (x, y) in self.coords.items():
            if math.hypot(event.x - x, event.y - y) <= 22:
                clicked_node = node
                break

        if clicked_node:
            if self.is_consistent(clicked_node, self.selected_color):
                self.assignments[clicked_node] = self.selected_color
                self.draw_graph()
            else:
```

```
                messagebox.showwarning(
                    "Constraint Violation",
                    f"Conflict detected! An adjacent linked node is already colored
{self.get_color_name(self.selected_color)}."
                )

    def get_color_name(self, hex_code):
        return self.color_names[self.colors.index(hex_code)]

    # --- Backtracking CSP Engine Loop Core ---
    def is_consistent(self, variable, value):
        for neighbor in self.topology.get(variable, []):
            if neighbor in self.assignments and self.assignments[neighbor] == value:
                return False
        return True

    def select_mrv_variable(self):
        unassigned = [v for v in self.nodes if v not in self.assignments]
        if not unassigned: return None

        def count_legal_values(var):
            return sum(1 for val in self.colors if self.is_consistent(var, val))

        return min(unassigned, key=count_legal_values)

    def backtrack(self):
        if len(self.assignments) == len(self.nodes):
            return True

        var = self.select_mrv_variable()
        if var is None: return False

        for value in self.colors:
            if self.is_consistent(var, value):
                self.assignments[var] = value

                # Dynamic refresh pipeline interface
                self.root.update()
                self.root.after(80)  # slightly faster calculation playback for 10 nodes
scale
                self.draw_graph()

                if self.backtrack():
                    return True

                del self.assignments[var]
                self.draw_graph()

        return False
```

```
        def run_ai_solver(self):
            self.assignments.clear()
            if self.backtrack():
                messagebox.showinfo("Solver Complete", "Success! The AI Backtracking
algorithm resolved all 10 nodes cleanly.")
            else:
                messagebox.showerror("Solver Error", "Failed to compute a valid mapping
solution.")

        def clear_assignments(self):
            self.assignments.clear()
            self.draw_graph()


    if __name__ == "__main__":
        main_window = tk.Tk()
        app = DynamicCSP10NodesGUI(main_window)
        main_window.mainloop()
```

Output:



Adversarial Source-code

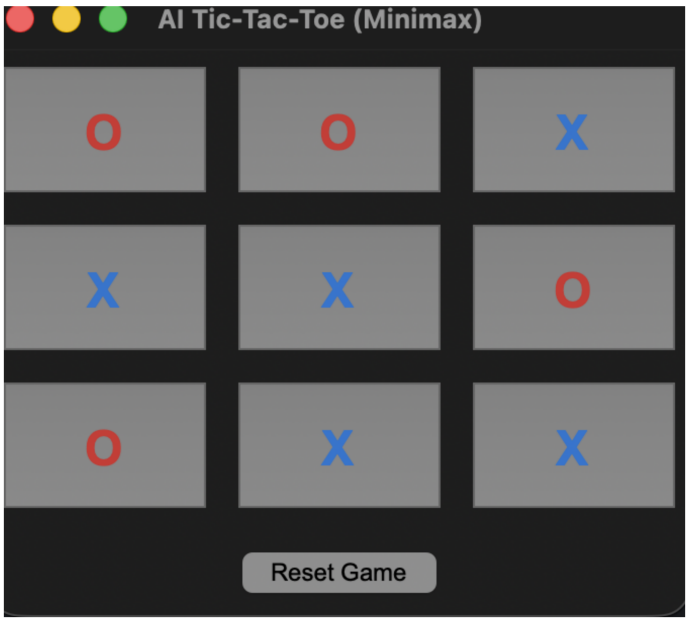
```
import math
import tkinter as tk
from tkinter import messagebox


class TicTacToeGUI:
    def __init__(self, root):
        self.root = root
        self.root.title("AI Tic-Tac-Toe (Minimax)")

        self.ai = 'O'
        self.human = 'X'
        self.board = [' ' for _ in range(9)]

        self.buttons = []
        self.create_widgets()
```

```
    def create_widgets(self):
        # Game grid configuration
        for i in range(9):
            # The lambda captures the current value of i as idx
            btn = tk.Button(
                self.root, text=' ', font=('Arial', 24, 'bold'),
                width=5, height=2, bg='#f0f0f0',
                command=lambda idx=i: self.human_move(idx)
            )
            # FIXED: Changed 'idx' to 'i' to match the loop iterator variable
            row = i // 3
            col = i % 3
            btn.grid(row=row, column=col, padx=5, pady=5)
            self.buttons.append(btn)

        # Reset Button
        reset_btn = tk.Button(
            self.root, text="Reset Game", font=('Arial', 12),
            command=self.reset_game, bg='#d9e2ec'
        )
        reset_btn.grid(row=3, column=0, columnspan=3, sticky="s", pady=10)

    def human_move(self, index):
        if self.board[index] == ' ' and not self.check_winner(self.ai) and not
self.check_winner(self.human):
            self.board[index] = self.human
            self.buttons[index].config(text=self.human, state='disabled',
disabledforeground='#1976D2')

            if self.evaluate_game_over():
                return

            # Trigger AI Turn immediately after a tiny delay
            self.root.after(300, self.ai_move)

    def ai_move(self):
        best_move = self.find_best_move()
        if best_move is not None:
            self.board[best_move] = self.ai
            self.buttons[best_move].config(text=self.ai, state='disabled',
disabledforeground='#D32F2F')
            self.evaluate_game_over()

    def find_best_move(self):
        best_score = -math.inf
        best_move = None
        for move in self.get_available_moves():
            self.board[move] = self.ai
            score = self.minimax(-math.inf, math.inf, False)
```

```
            self.board[move] = ' '
            if score > best_score:
                best_score = score
                best_move = move
        return best_move

    def minimax(self, alpha, beta, is_maximizing):
        if self.check_winner(self.ai): return 10
        if self.check_winner(self.human): return -10
        if ' ' not in self.board: return 0

        if is_maximizing:
            max_eval = -math.inf
            for move in self.get_available_moves():
                self.board[move] = self.ai
                evaluation = self.minimax(alpha, beta, False)
                self.board[move] = ' '
                max_eval = max(max_eval, evaluation)
                alpha = max(alpha, evaluation)
                if beta <= alpha: break
            return max_eval
        else:
            min_eval = math.inf
            for move in self.get_available_moves():
                self.board[move] = self.human
                evaluation = self.minimax(alpha, beta, True)
                self.board[move] = ' '
                min_eval = min(min_eval, evaluation)
                beta = min(beta, evaluation)
                if beta <= alpha: break
            return min_eval

    def get_available_moves(self):
        return [i for i, spot in enumerate(self.board) if spot == ' ']

    def check_winner(self, player):
        win_conditions = [
            [0,1,2], [3,4,5], [6,7,8], [0,3,6],
            [1,4,7], [2,5,8], [0,4,8], [2,4,6]
        ]
        return any(all(self.board[pos] == player for pos in cond) for cond in
win_conditions)

    def evaluate_game_over(self):
        if self.check_winner(self.human):
            messagebox.showinfo("Game Over", "Unbelievable! You won.")
            return True
        elif self.check_winner(self.ai):
            messagebox.showinfo("Game Over", "AI wins! Minimax is flawless.")
            return True
```

```
        elif ' ' not in self.board:
            messagebox.showinfo("Game Over", "It's a draw!")
            return True
        return False

    def reset_game(self):
        self.board = [' ' for _ in range(9)]
        for btn in self.buttons:
            btn.config(text=' ', state='normal', bg='#f0f0f0')


if __name__ == "__main__":
    main_window = tk.Tk()
    app = TicTacToeGUI(main_window)
    main_window.mainloop()
```

Output



Discussion

The experiment demonstrates how the Backtracking algorithm is effectively used to solve Constraint Satisfaction Problems (CSP) by systematically exploring all possible variable assignments. It shows that backtracking reduces unnecessary computations by pruning invalid solutions early when constraints are violated. The approach is also useful in adversarial problem scenarios where multiple decision paths must be evaluated. However, the method can become inefficient for large problem sizes due to its exponential time complexity.

Conclusion

The lab successfully illustrates the implementation of the Backtracking algorithm for solving CSP-based problems. It concludes that backtracking is a powerful technique for generating valid solutions by eliminating infeasible paths during the search process. Although it guarantees correctness, its performance depends on effective pruning strategies. Therefore, backtracking is best suited for problems with moderate input sizes and well-defined constraints.